ECO 310 - Problem Set 2
Due: Thursday, Feb 26

1. Income and Substitution Effects

   Assume perfect substitute preferences, with $u(x_1, x_2) = x_1 + x_2$.

   (a) Calculate the compensated demand bundle $(x_1^c(p_1, p_2, u^*), x_2^c(p_1, p_2, u^*))$. (Hint: you should have a piecewise function, depending on which good is cheaper).

   (b) Using your answer to (a), calculate the expenditure function, $e(p_1, p_2, u^*)$

   (c) Suppose that initially, $p_1 = 1$ and $p_2 = 2$ and $m = 10$, so that this consumer buys the bundle $(10, 0)$ and earns utility 10.

      i. Using your answer to (a), calculate the substitution effect if p_1 increases to 1.5.

      ii. Using your answer to (a), calculate the substitution effect if p_1 increases to 2.5.

2. Lagrangians. Suppose you use a Lagrangian to calculate the expenditure function in Q1b, so you set up

$$\mathcal{L} = -p_1 x_1 - p x_2 + \lambda_1 x_1 + \lambda_2 x_2 + \lambda_3(x_1 + x_2 - u^*)$$

   (a) Explain why this is the Lagrangian, eg what does each part of it represent.

   (b) Write out the optimality conditions, *if you guess that the utility constraint is binding, nonnegativity on good 1 is binding, and nonnegativity on good 2 is non-binding.*

   (c) Show that you get an invalid solution to (b) if $p_1 < p_2$. And explain why intuitively, the guess was wrong about which nonnegativity constraint binds.

3. Dave has preferences over lotteries which assign probabilities $p = (p_a, p_b, p_c)$ to three possible prizes: an apple, a banana, and a cherry. Suppose that Dave is indifferent between the lottery $p_1 = (1, 0, 0)$ and the lottery $p_2 = (0, \frac{1}{2}, \frac{1}{2})$ and that Dave strictly prefers the lottery $p_3 = (\frac{1}{2}, \frac{1}{2}, 0)$ to the lottery $p_4 = (0, \frac{3}{4}, \frac{1}{4})$. Show that Dave must violate at least one of the von Neumann Morganstern axioms (you don't need to figure out which one).

4. Suppose that with your driving style, you have chance p of getting a speeding ticket which costs you ϕ. Suppose further that the government wishes to hold the expected speeding ticket revenue $p\phi$ equal to a constant, $R \geq 0$. If you are strictly risk-averse, does a reduction in p (subject to the constraint that $p\phi$ be constant) help you or hurt you? (Suggested method: first write out EU, assuming initial wealth w_0. Differentiate in p, using $\phi = R/p$, to see if EU moves the same or opposite direction to p. Note that for a risk-averse consumer with concave u, we know that for any numbers $a < b$, $u(b) - u(a) < u'(a)(b - a)$).

5. Consider a decision-maker who expects to have a car accident with chance π; if this occurs, he will incur \$$L$ in damages. He can purchase as much auto insurance, q, as he likes, at a price of p per dollar of coverage: this means that if he pays pq upfront (as the insurance premium), he'll receive a payment of q from the insurance company **if an accident occurs**.

   (a) Write out his expected utility from purchasing insurance level q, assuming a utility-of-wealth function $u(w)$ and initial wealth w_0.

   (b) Show that the optimal level of insurance, q, solves

$$\frac{u'(w_0 - L + (1-p)q))}{u'(w_0 - pq)} = \frac{p(1-\pi)}{\pi(1-p)}$$

(c) Now assume that $u(w) = 1 - e^{-\alpha w}$, where $\alpha > 0$ (this is known as a "CARA", or "constant absolute risk aversion", utility function; α parametrizes risk aversion). Solve explicitly for the optimal insurance, and show that it does not depend on wealth.